1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

Dr Samit Bhattacharya
Computer Science and Engineering
IIT Guwahati

2 **L21 – Introduction to OOD**

Dr Samit Bhattacharya
Computer Science and Engineering
IIT Guwahati

3

4 **Recap**

- We covered requirement gathering stage
- We are discussing design stage

5

6 **Recap**

- We talked about two designs
  - Interface (& interaction)
  - Code

7 **Recap**

- The design-prototype-evaluate cycle primarily caters to interface design
- Once interface design stabilized and no further iteration required, we can focus on code design (can do in parallel also if no change in SRS)

8 **Design Concerns**

- Two issues
  - Where to start?
  - How to represent (design language)?

9 **Recap**

- Interface design – learned about guidelines as starting point

10 **Code Design - Recap**

- SRS – Software Specification Language

- Two phases
  - — Preliminary (high-level) design
  - — Detailed design (also called module-specification document)

11 **Basic Design Approaches - Recap**

- Function oriented – basic abstractions are functions

12 **Basic Design Approaches - Recap**

- Object oriented – basic abstractions are objects (instantiation of class)

13 **Design Representation - Recap**

- DFD (Data Flow Diagram) – for functional design approach
  - Already covered basics of DFD (and ERD)

14 **Recap**

- Today, we shall start on OOD & UML (to represent object-oriented design)

15 **OOD - Basics**

16 **Object Oriented Design Approach**

- Different way of thinking about a problem vis-à-vis procedural approach

17 **Example**

- Let us consider designing of a library management system

18 **Functional Design Approach**

- Procedural approach - design consists of a set of functions
  - We "think" of the system as provider of a set of "services" represented as "functions"

19 **Function (Service) Examples**

- Register – refers to the "service" of "registration to the system"
- Issue book – refers to the "service" of "issuance of book"
- Check availability – refers to the "service" of "knowing availability of particular book(s)"
- …

20 **Object Oriented Design Approach**

- In this approach, we no longer consider "services" offered by system
- Instead, we think of system as a collection of "objects" and their behavior (how they are related to each other)

21 **Example Revisited**

- Let's reconsider the library management system
- We can think of it as a collection of "objects" rather than a set of functions

22 **Example Objects**

- Member
- Book
- Book register
- …
- Also, how they are related to each other (e.g., member object can issue book object etc)

23 **Object vs Class**

24 **Class**

- Generic definition of objects
- Contains "attributes" & "operations" (also called methods/member functions)

25 **Attribute**

- The "data" part
- Indicates nature of data that an "object" can contain

26 **Operations**

- The "action" part
- Indicates the actions that can be performed with the data
  - Effectively denotes the behavior
- Can be *private* or *public*
  - Defines who can perform the operations

27 **Class Example**

28 **Object**

- Object - instantiation of class (assigning values to attributes and initialization of operations)

29 **Object Example**

30 **Another Class Example**

Name: Circle
Attribute(s): [can declare "private"]
double radius;
double centre_x_coordinate;
double centre_y_coordinate;

31 **Another Class Example**

Member function(s): [can declare "public"]
void setCenter (double x, double y);
void setRadius (double r);
void draw ();
double areaOfCircle ()

32 **Object for the Class**

- With Circle class definition, can create "Circle" objects
- Assign attribute values
- Set center, radius with the member functions
- Calculate area
- Draw it on screen

33 **Relevant Terms**

- OOA – object oriented analysis (SRS with objects instead of functions)
- OOD – object-oriented design
- OOP – object oriented programming (different than design)

34 **Reference**

- Rajib Mall (2018). Fundamentals of Software Engineering, 5th ed, PHI Learning Pvt Ltd. Chapter 7